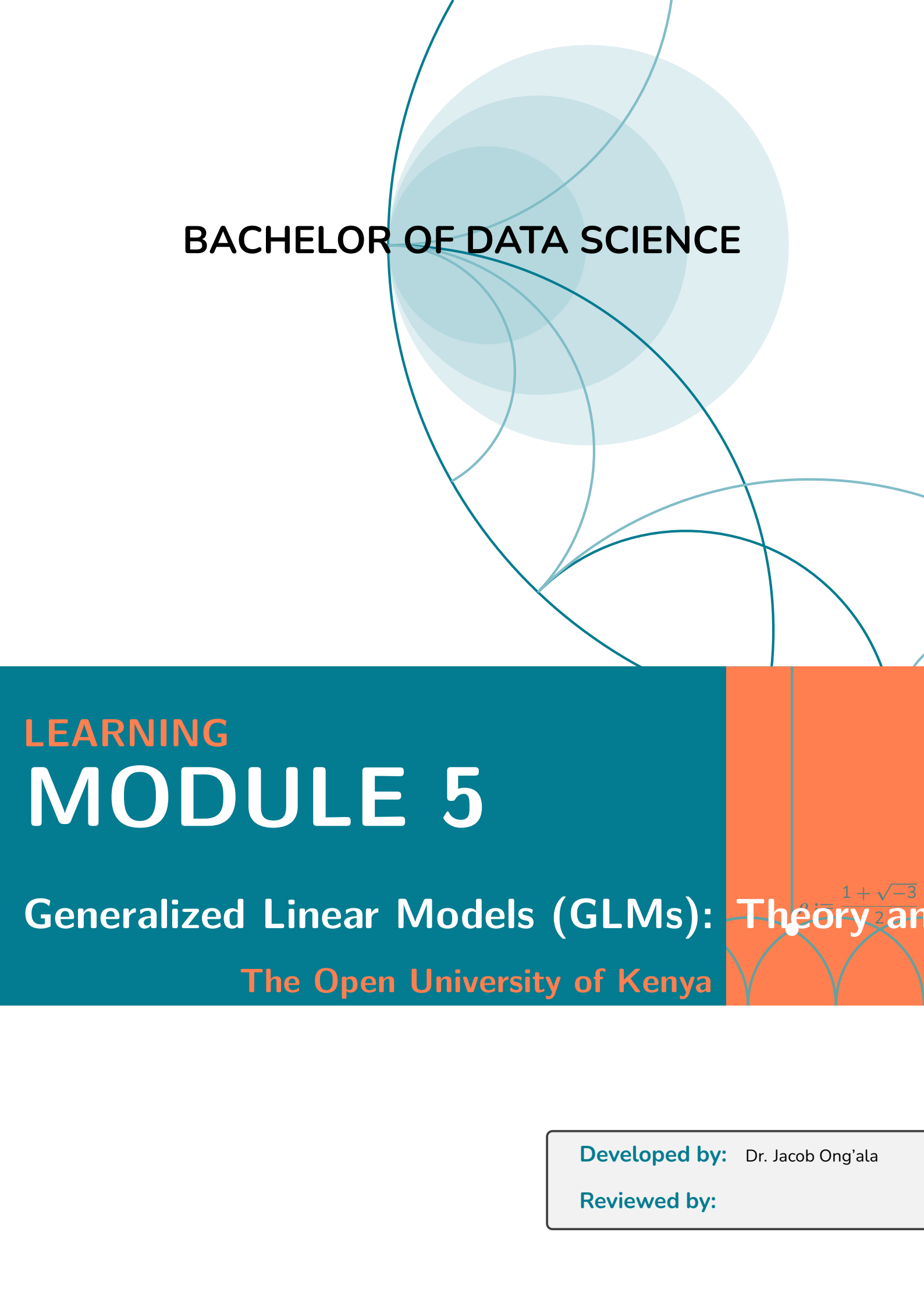




BACHELOR OF DATA SCIENCE

LEARNING MODULE 5

Generalized Linear Models (GLMs): Theory and Applications

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Reviewed by:

Programme Title	Bachelor of Data Science
Course Title	SST 304 - Linear Modeling
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Module 5 - Generalized Linear Models (GLMs): Theory and Framework

Instructional Hours: 3



Module Overview

This module introduces the foundational theory behind Generalized Linear Models (GLMs), a flexible extension of classical linear regression that accommodates various types of response variables. Students will explore how the exponential family of distributions underpins GLMs, the role of link functions, and the structure of GLMs in statistical modeling. Emphasis is placed on understanding the theoretical underpinnings that differentiate GLMs from ordinary linear models, setting the stage for practical implementation in real-world data analysis.

Module Learning Outcomes

By the end of this module, you should be able to:

1. Define the exponential family of distributions and identify common distributions that belong to it.
2. Explain the concept and purpose of link functions in GLMs.
3. Formulate a GLM using the structure $g(\mu) = X\beta$, and interpret its components.
4. Distinguish between canonical and non-canonical link functions.

Learning Activities

1. Read the lecture notes (PDF)
2. Read the assigned reference materials
3. Watch lecture videos
4. Complete the module assessment/Quiz



PRE ACTIVITIES

Review of Ordinary Linear Models (OLM) and Introduction to Generalized Linear Models

Instruction: Read the following summary and then complete the short task at the end.

The ordinary linear regression model is expressed as:

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i$$

Where:

- Y_i is the continuous response variable.
- x_{ij} are predictor variables.
- β_j are regression coefficients.
- $\varepsilon_i \sim N(0, \sigma^2)$ are independent errors.

Key Assumptions of OLM:

1. Linearity: The relationship between predictors and response is linear.
2. Normality: The errors ε_i are normally distributed.
3. Homoscedasticity: The variance of errors is constant.
4. Independence: Observations are independent of one another.

Limitations of Ordinary Linear Models and How GLMs Address Them

Limitation of Ordinary Linear Models (OLMs)	How Generalized Linear Models (GLMs) Address It
Assume errors are normally distributed	Allow for a range of distributions from the exponential family (e.g., Bernoulli, Poisson, Gamma) to model diverse types of response variables
Only suitable for continuous response variables	Support binary, count, proportion, and skewed continuous data
Assume constant variance (homoscedasticity)	Allow variance to be a function of the mean, accommodating heteroscedasticity
Use only the identity link function $\mu = X\beta$	Introduce flexible link functions (e.g., log, logit, inverse) to relate the mean response to the linear predictor
Cannot handle bounded outcomes like probabilities	Use appropriate link functions (e.g., logit) to ensure outputs remain within valid ranges such as $[0, 1]$
Inappropriate for modeling discrete outcomes	Incorporate discrete distributions such as Poisson and binomial
Assume additive error structure	Allow for different error structures, including multiplicative variance structures

Comparison of OLM Limitations and GLM Solutions



READING TASKS

Read the following Lecture notes and attempt the tasks that follows

1. Theory and Framework

Generalized Linear Models (GLMs) provide a unifying framework for modeling response variables that follow different distributions beyond the normal distribution, allowing for flexible modeling of binary, count, and skewed continuous data.

Generalized Linear Models extend OLMs to allow:

- Response variables from the **exponential family** (e.g., Bernoulli, Poisson, Gamma).
- A **link function** that relates the mean response to the linear predictor.
- Non-constant variance, modeled as a function of the mean.

1.1. Transition from Ordinary Linear Models to Generalized Linear Models

Ordinary Linear Model (OLM)

In a standard linear regression, we model the relationship between a continuous response variable Y_i and predictors x_i as:

$$Y_i = x_i^T \beta + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2)$$

Interpretation:

- $x_i^T \beta$ is the linear predictor or systematic component.
- The response Y_i is continuous and normally distributed.
- The expected value of Y_i is:

$$\mu_i = \mathbb{E}(Y_i) = x_i^T \beta$$

- The identity link is implicitly used: $g(\mu_i) = \mu_i$.

Limitations of the OLM Framework

OLM makes strict assumptions:

- The response variable is continuous and unbounded.
- The errors are normally distributed.
- Constant variance: $\text{Var}(Y_i) = \sigma^2$.

These assumptions do not hold for:

- Binary outcomes (e.g., success/failure).
- Count data (e.g., number of events).
- Skewed or bounded continuous data.

Generalizing the Model: GLM Framework

The **Generalized Linear Model (GLM)** allows us to model a broader class of response variables by introducing three modifications:

1. **Random Component:** Response Y_i follows a distribution from the exponential family (e.g., Bernoulli, Poisson, Gamma).
2. **Systematic Component:** As in OLM, the predictors form a linear predictor:

$$\eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$$

3. **Link Function:** A function $g(\cdot)$ links the mean response $\mu_i = \mathbb{E}(Y_i)$ to the linear predictor:

$$g(\mu_i) = \eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$$

Interpreting the GLM Structure

$$g(\mu_i) = \eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$$

Components:

- $\mu_i = \mathbb{E}(Y_i)$: Mean of the response variable.
- $g(\mu_i)$: Link function that transforms the mean to the linear predictor scale.
- $\eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$: Linear combination of predictors (same as in OLM).

Examples of Link Functions:

- **Identity link:** $g(\mu) = \mu$ — used in OLM and Gaussian GLMs.
- **Logit link:** $g(\mu) = \log\left(\frac{\mu}{1-\mu}\right)$ — for binary data.
- **Log link:** $g(\mu) = \log(\mu)$ — for count data (e.g., Poisson).
- **Inverse link:** $g(\mu) = 1/\mu$ — for Gamma-distributed responses.

Summary of Transition

- In OLM, $\mu_i = \mathbf{x}_i^T \boldsymbol{\beta}$ (identity link, normal errors).
- In GLM, the model becomes more flexible:

$$g(\mu_i) = \mathbf{x}_i^T \boldsymbol{\beta}$$

- The choice of g and the distribution of Y_i are tailored to the data.

A GLM consists of three components:

1. **Random Component:** The response variable Y_i follows a distribution from the exponential family, with mean μ_i .
2. **Systematic Component:** A linear predictor $\eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$, where $\mathbf{x}_i$ is a vector of predictors and $\boldsymbol{\beta}$ is the vector of unknown coefficients.
3. **Link Function:** A smooth monotonic function $g(\cdot)$ that relates the mean to the linear predictor:

$$g(\mu_i) = \eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$$

1.2. The Random Component of a Generalized Linear Model (GLM)

Definition

The **random component** specifies that the response variable Y is a random variable whose distribution belongs to the **exponential family of distributions**.

We observe n independent realizations:

$$Y_1, Y_2, \dots, Y_n$$

Each observation Y_i is modeled using a probability distribution from the exponential family.

The Exponential Family of Distributions

A probability distribution belongs to the **natural exponential family** if its probability density function (pdf) or probability mass function (pmf) can be expressed in the form:

$$f(y_i; \theta_i) = a(\theta_i) b(y_i) \exp [y_i Q(\theta_i)]$$

Interpretation of components:

- $a(\theta_i)$: A function depending only on the unknown parameter θ_i ; serves as a normalizing constant.
- $b(y_i)$: A function of the observed data point y_i ; independent of the parameter.
- $Q(\theta_i)$: A function of the parameter θ_i that appears in the exponent; often linear in canonical form.

Note: When there is existence of dispersion parameter, exponential family in a slightly different canonical form:

$$f(y_i; \theta_i, \phi) = \exp \left\{ \frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \phi) \right\}$$

Where:

- θ_i : Canonical (natural) parameter.
- $b(\theta_i)$: Cumulant function (ensures integration to 1).
- ϕ : Dispersion parameter.
- $c(y_i, \phi)$: A function ensuring the proper form of the distribution.



Example 1:

Poisson Distribution as an Exponential Class Variable

Suppose:

$$Y \sim \text{Poisson}(\lambda)$$

Then the probability mass function (pmf) is given by:

$$f(y; \lambda) = \frac{e^{-\lambda} \lambda^y}{y!}$$

We can rewrite this expression as:

$$f(y; \lambda) = e^{-\lambda} \cdot \frac{1}{y!} \cdot e^{\log \lambda^y}$$

$$f(y; \lambda) = e^{-\lambda} \cdot \frac{1}{y!} \cdot e^{y \log \lambda}$$

This matches the general form of the exponential family:

$$f(y_i; \theta_i) = a(\theta_i) \cdot b(y_i) \cdot \exp [y_i Q(\theta_i)]$$

Identifying components:

- $\theta = \lambda$
- $a(\lambda) = e^{-\lambda}$
- $b(y) = \frac{1}{y!}$
- $Q(\lambda) = \log \lambda$

Conclusion:

The Poisson distribution fits the form of the exponential family, and hence a Poisson random variable is an **exponential class variable**.



Example 2:

Identifying Exponential Family Components for the Normal Distribution

Consider the probability density function of a normal distribution with known variance σ^2 :

$$f(y; \mu) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(y - \mu)^2}{2\sigma^2} \right\}$$

Step 1: Expand the Exponent

$$-\frac{(y - \mu)^2}{2\sigma^2} = -\frac{y^2 - 2y\mu + \mu^2}{2\sigma^2} = -\frac{y^2}{2\sigma^2} + \frac{y\mu}{\sigma^2} - \frac{\mu^2}{2\sigma^2}$$

Thus, the density becomes:

$$f(y; \mu) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{y^2}{2\sigma^2} \right) \exp \left(\frac{y\mu}{\sigma^2} - \frac{\mu^2}{2\sigma^2} \right)$$

Step 2: Match to Exponential Family Form

The exponential family form is:

$$f(y; \theta) = a(\theta) b(y) \exp[y \cdot Q(\theta)]$$

- Natural (canonical) parameter: $\theta = \mu$
- $Q(\theta) = \frac{\mu}{\sigma^2}$
- $a(\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left(-\frac{\mu^2}{2\sigma^2}\right)$
- $b(y) = \exp\left(-\frac{y^2}{2\sigma^2}\right)$

Conclusion: The normal distribution with known variance belongs to the exponential family.

Examples of Distributions in the Exponential Family**QUIZ/QUESTIONS**

Identifying Exponential Family Components for the Bernoulli Distribution

Hint: Bernoulli distribution

$$f(y; \pi) = \pi^y (1 - \pi)^{1-y} = \exp\left\{y \log\left(\frac{\pi}{1 - \pi}\right) + \log(1 - \pi)\right\}$$

Each of these can be written in the form:

$$f(y_i; \theta_i) = a(\theta_i) b(y_i) \exp[y_i Q(\theta_i)]$$

Solution or the quiz**Bernoulli Distribution as a Member of the Exponential Family**

Suppose:

$$Y_i \sim \text{Bernoulli}(\pi)$$

where

$$P(Y_i = 1) = \pi \quad \text{and} \quad P(Y_i = 0) = 1 - \pi$$

Then, the probability mass function is:

$$f(y_i; \pi) = \pi^{y_i} (1 - \pi)^{1-y_i}$$

We manipulate the expression as follows:

$$f(y_i; \pi) = \pi^{y_i} (1 - \pi)^{1-y_i} = (1 - \pi) \left(\frac{\pi}{1 - \pi} \right)^{y_i} = (1 - \pi) \cdot \exp \left(y_i \log \left(\frac{\pi}{1 - \pi} \right) \right)$$

Matching with the Exponential Family Form

The general exponential family form is:

$$f(y_i; \theta_i) = a(\theta_i) b(y_i) \exp [y_i Q(\theta_i)]$$

Identified Components

- $\theta = \pi$
- $a(\pi) = 1 - \pi$
- $b(y_i) = 1$
- $Q(\pi) = \log \left(\frac{\pi}{1 - \pi} \right)$

Conclusion:

A Bernoulli random variable is a member of the exponential class.

1.3. Systematic Component of a Generalized Linear Model (GLM)

In the framework of Generalized Linear Models (GLMs), the systematic component refers to the part of the model that incorporates the explanatory variables into a linear predictor.

Definition

Let $\eta = (\eta_1, \eta_2, \dots, \eta_n)$ be the vector of linear predictors, where for each observation $i = 1, 2, \dots, n$, the predictor is defined as:

$$\eta_i = \sum_j \beta_j x_{ij} = \mathbf{x}_i^\top \boldsymbol{\beta}$$

Here:

- η_i is the linear predictor for the i -th observation,
- $\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{ip})^\top$ is the vector of predictor variables for the i -th observation,
- $\boldsymbol{\beta} = (\beta_1, \beta_2, \dots, \beta_p)^\top$ is the vector of regression coefficients.

Relation to Linear Regression

In classical linear regression, the expected value of the response variable is directly modeled as a linear combination of the predictors:

$$\eta_i = \mathbb{E}(Y_i) = \mathbf{x}_i^\top \boldsymbol{\beta}$$

However, this formulation restricts us to the assumption that the response variable Y_i must be continuous and normally distributed, which is often not realistic for many types of data (e.g., binary, count, proportion).

Generalization in GLMs

To overcome this limitation, GLMs introduce a link function $g(\cdot)$ where, $g(\cdot)$ be a monotonic and differentiable function and it relates the expected value of the response variable $\mu_i = \mathbb{E}(Y_i)$ to the linear predictor η_i :

$$\eta_i = g(\mu_i) \quad \text{or equivalently} \quad \mu_i = g^{-1}(\eta_i)$$

Here:

- $\mu_i = \mathbb{E}(Y_i)$ is the mean of the response variable,
- $g(\cdot)$ is a monotonic and differentiable link function,
- $g^{-1}(\cdot)$ is called the inverse link function.

In words, we are now modeling a function of the mean (or Expectation) as a combination of linear predictors.

Conclusion

The systematic component of a GLM allows for flexibility in modeling various types of response variables by connecting the linear predictor η to the expected response μ through a link function. This generalization extends the linear modeling framework to a broader class of distributions and response types.

1.4. Link Function

Link Function: A smooth, monotonic function $g(\cdot)$ that relates the mean of the response variable $\mu_i = \mathbb{E}(Y_i)$ to the linear predictor η_i :

$$g(\mu_i) = \eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$$

- The link function transforms the mean of the response variable so that it can be expressed as a linear function of the covariates.
- The choice of $g(\cdot)$ depends on the nature of the response variable and the distribution assumed.
- The function must be invertible: $\mu_i = g^{-1}(\eta_i)$.

Common Link Functions:

- **Identity link:** $g(\mu) = \mu$ (used in linear regression)
- **Logit link:** $g(\mu) = \log\left(\frac{\mu}{1-\mu}\right)$ (used in logistic regression)
- **Log link:** $g(\mu) = \log(\mu)$ (used in Poisson regression)
- **Probit link:** $g(\mu) = \Phi^{-1}(\mu)$ (used in binomial models with probit regression)

1.4.1. Canonical Link Function in Generalized Linear Models**Exponential Family Review**

Recall that a distribution belongs to the exponential family if it can be written in the form:

$$f(y_i; \theta_i) = a(\theta_i) b(y_i) \exp [y_i \cdot Q(\theta_i)]$$

Here:

- θ_i is the parameter of interest,
- $Q(\theta_i)$ is called the **natural parameter** (also referred to as the canonical parameter),
- $a(\theta_i)$ and $b(y_i)$ are functions depending only on the parameter and the observation, respectively.

Canonical Link Definition

In the context of a Generalized Linear Model (GLM), we model:

$$g(\mu_i) = \eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$$

where:

- $\mu_i = \mathbb{E}(Y_i)$ is the mean of the response variable,
- η_i is the linear predictor,
- $g(\cdot)$ is the link function.

The Canonical Link is the specific choice of the link function such that the transformed mean equals the natural parameter of the exponential family distribution. That is:

$$g(\mu_i) = Q(\theta_i)$$

This implies:

$$\eta_i = Q(\theta_i)$$

or equivalently,

$$\theta_i = g(\mu_i)$$

Interpretation and Importance

- The canonical link simplifies the estimation process and mathematical derivations.
- It often leads to desirable statistical properties such as sufficient statistics and simplified score equations.
- For each distribution in the exponential family, there exists a canonical link that aligns the linear predictor with the natural parameter.



Example 3:

Canonical Link for Bernoulli Distribution

Suppose $Y_i \sim \text{Bern}(\pi)$, where:

$$P(Y_i = 1) = \pi$$

$$P(Y_i = 0) = 1 - \pi$$

From earlier derivation, the Bernoulli probability mass function can be expressed in exponential family form as:

$$f(y_i; \pi) = (1 - \pi)(1) \cdot \exp \left(y_i \cdot \log \left(\frac{\pi}{1 - \pi} \right) \right)$$

This matches the exponential family structure:

$$f(y_i; \theta_i) = a(\theta_i) \cdot b(y_i) \cdot \exp (y_i \cdot Q(\theta_i))$$

with components:

- $Q(\pi) = \log \left(\frac{\pi}{1 - \pi} \right)$ — the natural (canonical) parameter,
- $a(\pi) = 1 - \pi$,
- $b(y_i) = 1$.

Therefore, the **canonical link function** is the **logit function**:

$$g(\pi) = \log \left(\frac{\pi}{1 - \pi} \right)$$

Setting this equal to the linear predictor gives:

$$g(\pi) = \mathbf{x}_i^\top \boldsymbol{\beta}$$

Thus, the canonical link for the Bernoulli (or Binomial) distribution is:

$$\log \left(\frac{\pi}{1 - \pi} \right) = \mathbf{x}_i^\top \boldsymbol{\beta}$$

which defines the logistic regression model.



QUIZ/QUESTIONS

Derive of the canonical link function for the Poisson distribution

Solution

Derivation of the Canonical Link: Poisson Distribution

Let $Y_i \sim \text{Poisson}(\lambda)$. The probability mass function (pmf) is given by:

$$f(y_i; \lambda) = \frac{e^{-\lambda} \lambda^{y_i}}{y_i!}, \quad y_i = 0, 1, 2, \dots$$

We aim to write this in the exponential family form:

$$f(y_i; \theta_i) = a(\theta_i) \cdot b(y_i) \cdot \exp(y_i \cdot Q(\theta_i))$$

Step 1: Rewriting the Poisson pmf

Start by expressing the pmf in terms of exponentials:

$$f(y_i; \lambda) = \frac{1}{y_i!} \cdot \exp(-\lambda) \cdot \exp(y_i \log \lambda) = [\exp(-\lambda)] \cdot \left[\frac{1}{y_i!} \right] \cdot \exp(y_i \log \lambda)$$

Step 2: Identify exponential family components

From the expression above, we can match the form:

$$f(y_i; \lambda) = a(\lambda) \cdot b(y_i) \cdot \exp(y_i \cdot Q(\lambda))$$

with:

- $Q(\lambda) = \log \lambda$ (natural/canonical parameter)
- $a(\lambda) = \exp(-\lambda)$
- $b(y_i) = \frac{1}{y_i!}$

Step 3: Define the canonical link

The canonical parameter is $Q(\lambda) = \log \lambda$. Since $\mu = \mathbb{E}(Y_i) = \lambda$, we define the canonical link function as:

$$g(\mu_i) = \log(\mu_i)$$

Step 4: Link function and linear predictor

The GLM assumes:

$$g(\mu_i) = \eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$$

Substituting the canonical link:

$$\log(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta}$$

or equivalently:

$$\mu_i = \exp(\mathbf{x}_i^\top \boldsymbol{\beta})$$

Conclusion: The **canonical link** for the Poisson distribution is the **log link**:

$$g(\mu) = \log(\mu)$$

This gives rise to the **log-linear model****, which is the standard form used in Poisson regression.

Examples of Canonical Links

Distribution	Mean (μ)	Canonical Link
Normal	μ	Identity: $g(\mu) = \mu$
Binomial	$\mu = \pi$	Logit: $g(\mu) = \log\left(\frac{\mu}{1-\mu}\right)$
Poisson	$\mu = \lambda$	Log: $g(\mu) = \log(\mu)$
Gamma	μ	Inverse: $g(\mu) = \frac{1}{\mu}$

Conclusion

The canonical link function connects the mean response μ to the natural parameter $Q(\boldsymbol{\theta})$ of the exponential family. Using the canonical link results in simplification of the likelihood equations and often improves the interpretability and computational efficiency of GLM estimation.

Why Use the Exponential Family in GLMs?

- It provides a ****unifying framework**** for many common distributions.
- It supports ****closed-form solutions**** for maximum likelihood estimation (MLE).
- It enables the use of a ****systematic modeling framework**** with link functions and linear predictors.
- It accommodates both ****discrete and continuous**** response variables.

Conclusion

In GLMs, the random component assumes that the response variable Y is generated from a distribution in the exponential family. This structure allows for flexibility in modeling various types of outcomes (binary, count, continuous, etc.) using a unified theoretical and computational approach.



CORE READING AND REFERENCES

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END OF MODULE ASSESSMENT

End of Module 5 Assessment

Attempt all the questions

Attempt All Questions

Q1.

The main motivation for introducing GLMs instead of ordinary linear models is:

- (a) To allow non-linearity in predictors.
- (b) To allow for non-normal distributions for the response variable.
- (c) To include random effects in the model.
- (d) To handle multicollinearity.

Q2.

Which of the following is **NOT** a distribution in the exponential family?

- (a) Bernoulli
- (b) Poisson
- (c) Exponential
- (d) Uniform

Q3.

Which of the following is the canonical link function for Poisson regression?

- (a) Identity link
- (b) Logit link
- (c) Log link
- (d) Inverse link

Q4.

Show that the Bernoulli distribution belongs to the exponential family and identify its natural parameter, $a(\pi)$, $b(y)$, and $Q(\pi)$.

Q5.

Derive the canonical link function for the Gamma distribution where the density is given as:

$$f(y; \mu, \phi) = \frac{1}{\Gamma\left(\frac{1}{\phi}\right)} \left(\frac{1}{\phi\mu}\right)^{\frac{1}{\phi}} y^{\frac{1}{\phi}-1} \exp\left(-\frac{y}{\phi\mu}\right)$$

End of Module 5 Assessment-Rubric

Q1. Answer: b)

GLMs allow for non-normal response distributions (exponential family).

Q2. Answer: d)

Uniform distribution does not belong to the exponential family.

Q3. Answer: c)

For Poisson distribution, the canonical link is the log link.

Q4. For the Bernoulli distribution:

$$f(y; \pi) = \pi^y (1 - \pi)^{1-y}$$

Rewrite as:

$$f(y; \pi) = (1 - \pi) \cdot \left(\frac{\pi}{1 - \pi} \right)^y = (1 - \pi) \cdot \exp \left(y \log \left(\frac{\pi}{1 - \pi} \right) \right)$$

Thus, matching the exponential family form:

$$f(y; \theta) = a(\theta) b(y) \exp(yQ(\theta))$$

Identifying:

$$Q(\pi) = \log \left(\frac{\pi}{1 - \pi} \right)$$

$$a(\pi) = 1 - \pi$$

$$b(y) = 1$$

The natural parameter is: $Q(\pi)$.

Q5. For the Gamma distribution:

We rewrite the given density:

$$f(y; \mu, \phi) = \exp \left\{ -\frac{y}{\phi\mu} + \left(\frac{1}{\phi} - 1 \right) \log y - \log \Gamma \left(\frac{1}{\phi} \right) - \frac{1}{\phi} \log(\phi\mu) \right\}$$

Expressing in exponential family form:

$$f(y; \theta) = \exp \left\{ \frac{-y}{\phi\mu} - \frac{1}{\phi} \log \mu + \text{terms not depending on } \mu \right\}$$

Thus, identifying:

$$Q(\mu) = -\frac{1}{\mu}, \quad \text{so canonical link is } g(\mu) = \frac{1}{\mu}$$